

1. We have gone through several kinds of equations now and let's sum up some of these *as proportions*:

- acceleration is _____ net force.
- assuming constant acceleration and beginning at rest, an object's velocity is _____ the displacement.
- assuming constant acceleration and beginning at rest, an object's displacement is _____ the elapsed time.
- for an object that has been dropped, the distance it has fallen is _____ its velocity at that distance.

2. I push a 100 kg box starting at rest along a friction-less floor, with a force of 100 N for 10 s. How fast is the box going at this point? If I did the same thing to a 200 kg box, then how fast is it going after 10 s?

3. I push a 100 kg box starting at rest along a friction-less floor, with a force of 100 N over a distance of 10 m. How fast is the box going at this point? If I did the same thing to a 200 kg box, then how fast is it going after 10 m?

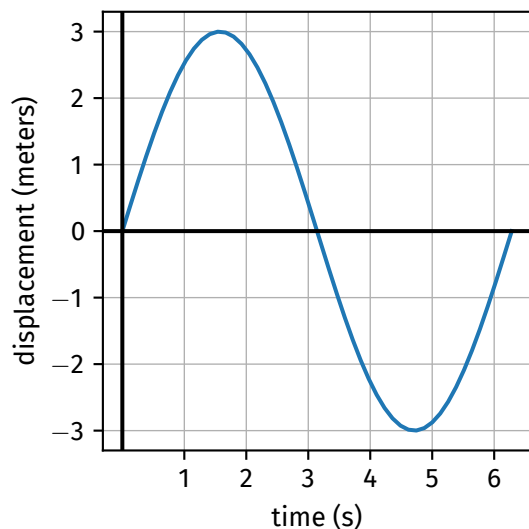
Some starters:

- What is the net force on the box?
- What is the acceleration of the box?
- What is the final velocity after 10 m

4. Following up on the previous problem, if I stopped pushing after 10 m and the box continued with its speed, and then at then started sliding up a 20° ramp, then how far along the length of the ramp would the box rise? What height is this above the horizontal? Do the 100 kg and the 200 kg box rise to the same height?

Some starters:

- What is the net force on the box as it goes up the inclined plane?
- What is the acceleration of the box as it goes up the inclined plane?
- What is the sign of the displacement of the box going up the plane?
- Is the sign of acceleration the same or different than displacement?



8. If a soccer ball with a radius of 10 cm is rolls along the ground without slipping at 5 m/s, then how many revolutions does it roll through in 10 s and what distance has a point on the edge of the ball traveled? Some starters:

- How fast is it *spinning*? By that we mean *angular speed*.
- How many radians does the ball rotate through in this time? What is that in revolutions?
- How far does it roll in this time? Is this the same distance as the distance of a point on the edge of the ball? Why or why not?

9. Following up on the previous problem, how many seconds does it take for the ball to complete one revolution? This amount of time is referred to as the *period* of its rotation, and this is a similar characteristic time for the motion of the ball as the *period of a pendulum* was in the first lab.
10. Another follow up. How many revolutions does the ball travel through *per second*? You could convert this from angular speed ω that you would have calculated in the first instance of this problem, but if all you new was the period of the ball's rotation, how could you calculate it from there? (*Hint: what is the difference between revolutions per second and seconds per revolution?*) This quantity of revolutions per unit of time is sometimes called *frequency*.

